

LUIS SAENZ

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<https://luisasaenz.github.io/portfolio/>

PROFESSIONAL SUMMARY

Robotics Engineering graduate at Arizona State University and US Army veteran with hands-on experience in PCB schematic design, analog and digital circuit design, and electromechanical prototyping. Led hardware development for a ceiling-mounted drone docking system, delivering a functional electromechanical prototype from concept through integration testing. Brings proven leadership, technical documentation user and creator, and systems-level thinking from four years of active military service.

WORK EXPERIENCE

Information Technology Specialist | *United States Army Reserves*

Oct 2020 – Jun 2023

7453rd Medical Backfill Battalion

- Installed, operated, and maintained military computer systems, networks, routers, switches, and end-user devices.
- Diagnosed and resolved hardware and software issues while ensuring compliance with Army cybersecurity standards.
- Provided technical support and training to unit personnel on military computer systems and networks.

Aerial Delivery Specialist | *United States Army*

Feb 2016 – Sep 2020

82nd Airborne Division

- Managed and supervised multiple teams of up to 10 personnel daily. Delivered all cargo without malfunctions or operational failures, saving the organization more than \$100,000 per vehicle over two years.
- Led and trained 20 riggers in over 400+ hours on how to properly engineer aerial equipment, manage clients, and conduct proper safety procedures.
- Packed 600+ cargo and personnel parachutes across routine and mission-critical operations, where a single packing error means mission failure and equipment loss, demonstrating the attention to detail and precision required in a zero-defect environment.

Software Developer Student | *Microsoft Software & Systems Academy*

Jan 2020 – May 2020

- Wrote and tested database queries using T-SQL in SQL Server Management Studio 18, building the ability to retrieve, filter, and manipulate data stored in relational databases.
- Applied OOP principles and programming algorithms to build console-based applications in C# (.NET Core) using Visual Studio 2019, dedicating 25–30 hrs/week to accelerate proficiency from zero baseline.
- Used Git, GitHub, and Fossil to save, track, and share code changes while learning the industry standard workflow for managing code across solo and team projects.

PROJECTS

Ceiling-Mounted Drone Docking System | *Arizona State University — (Axon Sponsor) 2025 – 2026*

Electrical & Firmware Lead | Cross-functional team of 4 engineers

- Designed complete electrical schematics and constructed a PCB in Cadence. Generated Gerber files (top, bottom, outline, top and bottom soldermask), checked manufacturability with JLCPCB's DFM tool, and ordered the prototype board. The fabricated PCB enabled fast assembly and speed up electrical testing.
- Developed firmware to control door movement, locking, and the undocking sequence through servo motors and actuators, ensuring components operate in the correct order and reducing mechanical failures during testing.
- Integrated a remote server interface into the docking system, enabling wireless command input and status monitoring from outside the immediate operating environment.

- Analyzed test data to identify failure modes (harness separation, PLA axle torsion, gear misalignment), implemented targeted fixes (added harness prongs, replaced the PLA rod with a D-shaft, repositioned gears, and added a second servo per door), and re-tested. Results of fixes are consistent door-operation times and achieving 100% undocking reliability.
- Co-authored the test plan, defined clear pass/fail criteria and step-by-step procedures, and produced concise test reports summarizing outcomes and recommended next steps to guide each design iteration.
- Authored electrical schematics, test reports, and interface documentation (pinouts, wiring, signal and protocol details) and maintained the BOM to support prototype fabrication and assembly.
- Led embedded hardware integration; implemented power distribution logic, microcontroller interfaces, and actuator sequencing to ensure reliable system operation during integration testing.

EDUCATION

Bachelor of Science — Robotics Engineering | Arizona State University

May 2026

Polytechnic Campus, Mesa, AZ

- Relevant coursework: electrical circuits, electromechanical, embedded systems, robotic kinematics & dynamics, systems modeling, engineering statistics, programming
- GPA: 3.3/4.0

Microsoft Software & Systems Academy | Embry-Riddle Aeronautical University

May 2020

Advanced technical course in cloud application development

TECHNICAL SKILLS

PCB & Hardware Design: KiCad, Cadence

Programming Languages: C++, Python, C#, MATLAB, T-SQL, HTML/CSS

Tools, Instruments & Platforms: Raspberry Pi, Git/GitHub, LTSpice, Microsoft Visual Studio, SQL Server Management Studio, Multimeter, bench power supply, oscilloscope, FreeCad (parametric 3D-modeling)

CERTIFICATIONS

MTA 98-361 — Software Development Fundamentals

Nov 2019

Querying Data with Transact-SQL

Jun 2020